

Python Certification Program

1. Course Scope and Intent

Topic	Python for Data Automation, Analysis, and Implementation
Purpose	Enable participants to gain practical Python skills, deliver solutions, and/or teach Python professionally
Intended Output / Skill	Write maintainable scripts, automate workflows, implement scalable solutions, teach/mentor others
Target Audience	Beginners Professionals SMEs

2. Program Framework

Element	Implementation
<i>Experts</i>	Senior Python developers, data engineers, certified trainers, industry consultants
<i>Duration</i>	10-25 hours total, with 10-15 min micro-sessions
<i>Platform</i>	Online LMS (e.g. Moodle) + optional live workshops for labs or Q&A
<i>Format</i>	Mix of self-paced videos, interactive coding labs, scenario exercises, and live sessions
<i>Course Evaluation</i>	Participant feedback, retention metrics, engagement tracking, sign-up rates, exam pass rates

3. Curriculum & Learning Outcomes (Based on Bloom's Taxonomy)

<i>Learning Path</i>	<i>Audience</i>	<i>Bloom's Level</i>	<i>Learning Outcomes</i>	<i>Assessment</i>
Foundation	No Coding Experience	Remember > Understand	Recall Python syntax, data types, control flow; explain workflows; identify standard libraries; navigate dev environments	Quizzes, guided coding exercises
Practitioner	Developers / Analysts	Apply > Analyze	Apply Python to real problems; implement scripts; debug code; analyze performance	Hands-on projects, scenario exercises, code review
Partner Certification	Trainers / Consultants	Analyze > Evaluate > Create	Design scalable solutions; evaluate design choices; create end-to-end implementations; deliver training	Real world example project, recorded demo
Expert	SMEs	Create	Create reference architectures and reusable frameworks; mentor learners; contribute to curriculum	Peer-reviewed contributions, expert panel evaluation, live demos

4. Industry-Specific Requirements

Main Requirements	Core Python programming and best practices
	Problem-solving and data automation
	Scalable solution design and architecture
	Debugging and optimization
	Technical documentation and communication
	Customer-facing delivery for partners
	Mentorship and enablement at expert levels

5. Learning Materials

Format	Implementation
Self-paced	Pre-recorded video lessons, interactive notebooks, downloadable code samples
Instructor-Led	Live workshops, webinars, Q&A sessions
Hands-On Labs	Practice exercises, coding challenges, mini-projects
Reference Materials	Solution templates, example code, documentation guides

6. Assessment Methods & Certification

<i>Assessment Type</i>	<i>Bloom Level</i>	<i>Example</i>	<i>Scoring / Weight</i>
Knowledge Quiz	Remember / Understand	Multiple-choice questions on syntax & libraries	20-30%
Hands-On Project	Apply / Analyze	Automate a dataset processing task	25-30%
Scenario-Based Task	Analyze / Evaluate	Optimize a script or workflow	15-20%
Real world example Project	Evaluate / create	End-to-end Python solution with tests & documentation	20-25%
Partner Delivery Assessment	Create	Recorded or live teaching/demo session	10-15%

Passing Thresholds:

- Practitioner: 70%
- Partner Certification: 80%
- Expert: 90%

7. Certification Rubric / Grading Matrix

<i>Criterion</i>	<i>Practitioner</i>	<i>Partner</i>	<i>Expert</i>
Technical Correctness	40%	30%	25%
Architecture & Design	20%	25%	30%
Problem Solving & Analysis	25%	20%	15%
Documentation	15%	15%	15%
Delivery	-	10%	15%

References

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